Zhenyuan Zhang

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Education

PhD candidate in Human Movement Biomechanics Jul 2023 – May 2027(Expected)
Research Institute for Sport and Exercise Science
Liverpool John Moores University, United Kingdom

Master of Science in Sport and Clinical Biomechanics Sep 2020 – Nov 2021
School of Sport and Exercise Science
Liverpool John Moores University, United Kingdom

Bachelor of Education in Human Movement Science Sep 2016 – Jun 2020
School of Exercise and Health Science
Chengdu Sport University, China

Skills

Programming Stacks: Python, Matlab, Git, GitHub, Bash, LaTeX

Machine Learning: PyTorch, TensorFlow, Scikit-learn, ONNX

Cloud Computing: Slurm, AWS, Azure

Biomechanics Tools: Optical Motion Capture, Inertial Measurement Units, Force Plates, Electromyography

Computer Simulations: OpenSim, Visual3D, MyoSuite

Experiences

Research Engineer - Machine learning & Biomechanics Jun 2026 – Present
Uplyft Wearables Ltd., UK | www.uplyftwearables.com

I work as a research engineer in machine learning and biomechanics, building intelligent wearable products for real-world applications that remotely monitor and assess rehabilitation progress, daily human gait, and athletic performance used by clinicians, athletes and coaches.

- Developing pipelines for training and validating machine learning models that learn human movement from IMU and pressure sensors. Deploying them with framework-agnostic ONNX runtimes to a real-world wearable intelligent products.
- Developing an internal Python package "Model-Factory" that builds mutiple machine learning models at scale. It provides a unified interface for model development and deployment, automates model construction for downstream applications.
- Developing another internal Python package "Biomechanics-Engine" as a comsumer of machine learning models, with integration of the physics-based computation, powering all biomechanics-related data processing in both real-time and offline for motion tracking, gait analysis, and performance evaluation.
- Building cloud infrastructure with GitHub and Microsoft Azure for software shipping and production. This integrates multiple services, including serverless Neon PostgreSQL Database, Azure Blob Storage, Azure Virtual Machines, and Azure Machine Learning workspace into CI/CD workflows that processes big data (>1TB) at scale.

Machine Learning and Biomechanics Researcher Jul 2023 – Jun 2026
SportScientia Ltd., UK | www.sportscientia.com

In this PhD-Industry partnership, I'm developed deep learning neural networks deployed on AWS to estimate biomechanical loading from instrumented insoles during real-world sports activities for high-performance athletes.

- Developed deep learning models to estimate ground reaction forces from wearable insoles instrumented with IMUs and pressure sensors during various human movements.
- Validated wearable insoles with sensor fusion algorithms for measuring spatiotemporal gait parameters against optical motion capture and force plates for athletic performance and load monitoring.
- Assisted in developing cloud computing infrastructures on AWS to deploy machine learning models and automate data processing and analysis for the wearable system.

Biomechanics Laboratory Technician Sep 2025 – Jan 2026

Liverpool John Moores University, UK

I built and fine-tuned 3 advanced biomechanics systems with the senior technician, provided professional training for staff and students. Also responsible for managing them for both teaching and research activities.

- System1: 10 Qualysis® Arqus motion capture cameras integrated with 8 Qualysis® Miqus cameras for marker-less motion capture, 2 Kistler® force plates, 16 Delsys® Trigno EMGs.
- System2: 8 Qualysis® Arqus motion capture cameras integrated with a Treadmetrix® treadmill (AMTI® force plate embedded), 16 Delsys® Trigno EMGs and 8 Noraxon® EMGs.
- System3: 14 Vicon® Vero motion capture cameras integrated with 16 Vicon® T-series cameras, 2 Kistler® force plates, 8 Vicon® Blue Trident IMUs and 16 Delsys® Trigno EMGs.

Graduate Research Assistant Nov 2021 – Jul 2023

Liverpool John Moores University, UK

- Collaborated with New Balance Athletics, USA to test biomechanical interactions between soccer boots and artificial turfs with different studs configurations using high-speed motion capture and force plates.
- Collaborated with New Balance Athletics, USA to test effects of different running shoe configurations on running mechanics and muscle co-contractions during treadmill running using motion capture, instrumented treadmill and EMGs.

Projects

TechLayer | GitHub

- A Python project implemented a deep learning model to predict Ground Reaction Forces (GRFs) from IMU and pressure sensor data collected from instrumented insoles during various dynamic sports movements.
- The model is trained and validated on a dataset of 32 participants using High-Performance Computing (HPC) clusters and demonstrated high accuracy in predicting vertical and anterior-posterior GRFs against force plates.

Wearable_IK | GitHub

- A Python project integrating open-source sensor fusion algorithms and OpenSim API to estimate joint kinematics from IMU data and validating the results against optical motion capture data.
- It automates the entire workflow from data loading, preprocessing, sensor fusion, sensor-to-segment calibration, solving inverse kinematics, to results visualization
- It also features parallelized computing to speed up processing and a calibration-free approach for IMU sensors as it does not require magnetometers.

Wearable_System | GitHub (Under Development)

- A Python project integrating **TechLayer** and **Wearable_IK** to generate muscle-driven simulations of human movement while solving optimal control problems using wearable sensor (instrumented insoles + IMUs) data with OpenSim Moco. It also aims to validate the simulation results against traditional biomechanics measurement from optical motion capture, force plates and EMGs.

My_Website | GitHub

- A project to build and deploy my personal academic website from an open-source TypeScript template for fun.

Publications

Zhang, Z., Verhuel, J., Robinson, M., and Lake, M., Magnetometer-free 3D inertial motion capture for slow and fast human movements: validation and practical recommendations. Journal of Biomechanics, Under Review. 2026

Zhang, Z., Robinson, M., Verhuel, J., and Lake, M., Estimating ground reaction forces using instrumented insoles with deep learning from daily to sports-related movements. IEEE Transactions on Biomedical Engineering, Under Review. 2026

Zhang, Z., Verhuel, J., Robinson, M., and Lake, M., Estimating ground reaction forces in dynamic sports movements using instrumented insoles and deep learning. Oral Presentation at 30th Congress of International Society of Biomechanics, p.85033, Stockholm, Sweden. 2025

Yang, C., Yang, Y., Xu, Y., **Zhang, Z.**, Lake, M., and Fu, W. Whole leg compression garments influence lower limb kinematics and associated muscle synergies during running. Frontiers in Bioengineering and Biotechnology, 12, 1310464. 2024

Zhang, Z. and Lake, M. Rate of knee flexion at the instant of landing during running can influence initial knee joint stiffness estimates due to running shoe cushioning. Oral Presentation at 29th Congress of International Society of Biomechanics, p.314, Fukuoka, Japan. 2023

Zhang, Z. and Lake, M. A re-examination of the measurement of foot strike mechanics during running: the immediate effect of footwear midsole thickness. Frontiers in Sports and Active Living, 4, 824183. 2022

Zhang, Z. and Lake, M. A comparison of unmatched and matched filtering approaches for knee joint stiffness calculation during running. Oral Presentation at 40th International Society of Biomechanics in Sports, Proceedings Archive, 40(1), 807, Liverpool, United Kingdom. 2022

Contributions

ISB Congress Session Moderator: Chaired the parallel session "Wearable Measurement Devices I" at the 30th Congress of the International Society of Biomechanics (Stockholm, Sweden, 2025).

Teaching

Graduate Teaching Assistant Nov 2021 – Nov 2025

Liverpool John Moores University, UK

- Assisted in biomechanics lectures and practical laboratory sessions for optical motion capture cameras, force plates, IMUs and EMGs at undergraduate and master levels.
- Provided one-on-one tutorial support for students' projects and technical training of setting up and operating the equipment for data collection of students' projects.

Scholarships

University-Industry Matched PhD Fund: A three-year matched funding package awarded jointly by Liverpool John Moores University and industry partner to support collaborative PhD research. (Jul 2023)

LJMU International Achievement Scholarship: A scholarship awarded to international students with excellent academic performance to pursue postgraduate studies at Liverpool John Moores University. (Sep 2020)

References

Prof. Mark Lake, Professor of Biomechanics | M.J.Lake@ljmu.ac.uk

Dr. Mark Robinson, Associate Professor of Biomechanics, | M.A.Robinson@ljmu.ac.uk

Prof. Gabor Barton, Professor of Clinical Biomechanics, | G.Barton@ljmu.ac.uk